Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)(i)	48	**1**	
1(a)(ii)	$\frac{5}{16}$ cao	**1**	**FT** their **(a)(i)** provided simplification required and answer given in its simplest form
1(a)(iii)	44 nfww	**2**	**M1** for $\frac{28}{7}[\times 11]$ oe
1(b)(i)	120	**1**	
1(b)(ii)	116	**2**	**M1** for at least first four or last four correctly ordered
1(b)(iii)	110	**2**	**M1** for $\frac{123+98+116+45+67+165+156}{7}$ oe
1(c)(i)	68	**2**	**M1** for $\frac{4.2-2.5}{2.5}[\times 100]$ oe or $\frac{4.2}{2.5}\times 100[-100]$ oe or $\left(\frac{4.2}{2.5}-1\right)[\times 100]$ oe
1(c)(ii)	14 100 or 14 140 or 14 137 to 14 139	**2**	**M1** for $\frac{4}{3}\times\pi\times 15^3$ oe
	cm^3	**1**	
1(d)	*A* With correct comparisons made of the 3 bottles with suitable accuracy shown	**3**	**M2** for 3 correct comparable values, or for a correct method to compare 3 bottles shown but not evaluated to enough accuracy or **M1** for 2 correct comparable values or for a correct method to compare 3 bottles but not evaluated
2(a)(i)	1.5 4 –4 –1.2	**3**	**B2** for 3 correct or **B1** for 1 or 2 correct
2(a)(ii)	Correct graph drawn	**4**	**B3FT** for 12 or 11 correct plots **B2FT** for 10 or 9 correct plots **B1FT** for 8,7 or 6 correct plots

Question	Answer	Marks	Partial Marks
2(a)(iii)	2	**1**	
2(a)(iv)	$y = x$ oe $y = -x$ oe	**2**	**B1** for each
2(a)(v)	Correct ruled line	**1**	
2(a)(vi)	-2.4	**1**	**FT** *their* graph and $y = 2.5$
2(b)	Correct ruled line	**1**	
2(c)	$[y =]3x + 4$ cao	**2**	**B1** for final answer $[y =]3x + k$, $k \neq 5$ or $[y =]jx + 4$, $j \neq 0$
3(a)(i)	Obtuse	**1**	
3(a)(ii)	113	**1**	
3(b)(i)	Sketch of a rhombus	**1**	
3(b)(ii)	Rhombus cao	**1**	
3(b)(iii)	70, 110, 110	**2**	**B1** for 110 or **M1** for (360 – 70 – 70) ÷ 2 oe or **M1** for $70 + 70 + x + x = 360$ oe soi
3(c)(i)	Angles [in a] triangle add to 180	**1**	
3(c)(ii)	$x + 6y = 180$ oe	**1**	
3(c)(iii)	Correctly eliminating one variable	**M1**	**FT** *their* **(c)(ii)**, if linear in x and y
	$[x =]72$	**A1**	
	$[y =]18$	**A1**	If **M0** scored, **SC1** for 2 values satisfying one of the original equations or *their* equations in **(c)(ii)** **SC1** if no working shown but 2 correct answers given
4(a)(i)	39	**2**	**M1** for 7.8×5 or **B1** for 7.8 and 5 marked on two correct sides of rectangle A
4(a)(ii)	5 7.8 12	**2**	**M1FT** for $\frac{468}{7.8 \times 5}$ oe or for $\frac{468}{\textit{their}\ \textbf{(i)}}$
4(b)	768π final answer	**2**	**M1** for $\pi \times 8^2 \times 12$ oe